Sek 5/11

SHRI SHIKSHAYATAN SCHOOL

PRE-BOARD EXAMINATION : 2025-26

Subject – **SCIENCE (086)**

Class–X

Time allowed : 3 hrs. **Max. Marks : 80**

Set A

Name ____________________

Class X Section A Roll No. 10

General Instructions :

(i) This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

(ii) All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.

iii) Students will divide the answer book in the above 3 sections.

iv) Replies of questions are to be written only within the space identified for the concerned section only.

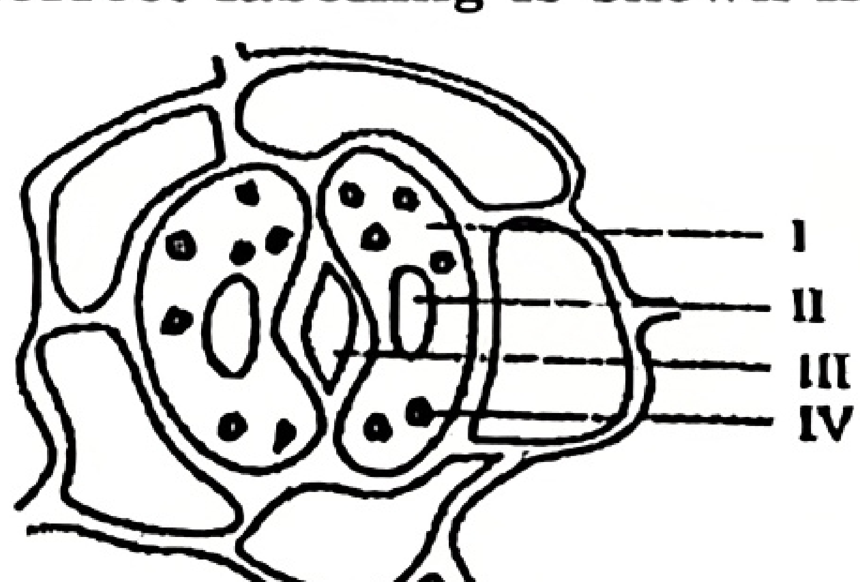
v) Reply cf a section should not be written or mixed in any other section.

vi) In case, if replies are mixed, these will not be evaluated.

SECTION – A

Choose the correct option :

1. In the sketch of stomatal apparatus given alongside, the parts I, II, Ill and IV were labelled differently by four students. The correct labelling is shown in : [1]



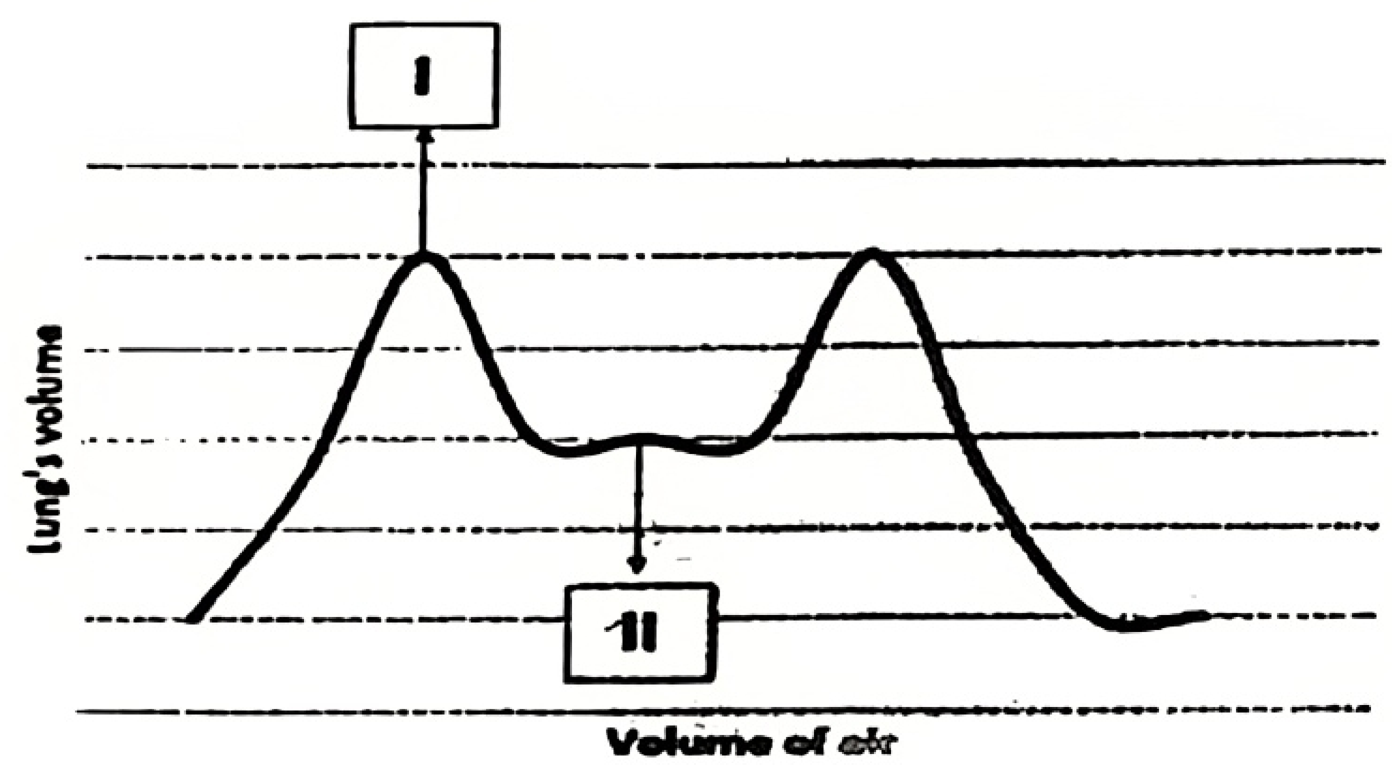
a) (I) guard cells, (II) stoma, (III) starch granule, (IV) nucleus

b) (I) cytoplasm, (II) nucleus, (III) stoma, (IV) chloroplast

c) (I) guard cells, (II) starch, (III) nucleus, (IV) stoma

d) (I) cytoplasm, (II) chloroplast, (III) stoma, (IV) nucleus

2. Given below is the graphic illustration of the change in the volume of the lungs during the process of breathing-



Which of the given steps during breathing explains the change from I to II? [1]

a) The increase in the thoracic cavity.

b) The contraction and downward movement of the diaphragm.

c) The decrease in the thoracic cavity.

d) The movement of the rib cage upward and outward.

3. Which of the following statements are true about the brain? [1]

(i) The main thinking part of the brain is hind brain.

(ii) Centres of hearing, smell, memory, sight, etc. are located in fore brain.

(iii) Involuntary actions like salivation, vomiting, blood pressure are controlled by the medulla in the hind brain.

(iv) Cerebellum does not control posture and balance of the body. [1]

(a) (i) and (ii) (b) (i), (ii) and (iii)

(c) (ii) and (iii) (d) (iii) and (iv)

4. A growing seedling is kept in a dark room. A burning lamp is placed near to it for a few days. The top part of seedling bends towards the burning candle. This is an example of 1

(a) Chemotropism (b) Hydrotropism

(c) Phototropism (d) Geotropism

5. Which of the following steps correctly show how traits are inherited in sexually reproducing organisms? [1]

i) DNA in the parent cell is copied before reproduction

ii) Male and Female gametes are formed, each having only half the amount of DNA as compared to the parent cell

iii) Male and female gametes fuse during fertilisation to form a zygote.

iv) The zygote gets DNA from both parents, resulting in variation

v) Genes present in DNA decide the traits in the offspring

a) i), iii), iv) b). i), ii), iii), iv), v)

c) ii), iii), iv) d) iii), iv), v)

6. Some waste products are listed below :

Grass Cuttings, Polythene Bag, Plastic Toys, Used Tea Bags, Old Clothes, Paper Straw

Which group of waste materials can be classified as non-biodegradable? [1]

a) Plant waste, used tea bags

b) Polythene bags, plastic toys

c) Used tea bags, paper straw

d) Old clothes, broken footwear

7. Which statement shows the interaction of an abiotic component with a biotic component in an ecosystem? [1]

a) A grasshopper feeding on a leaf

b) Rainwater running down into the lake

c) An earthworm making a burrow in the soil

d) A mouse fighting with another mouse for food

Following questions consist of two statements – Assertion (A) and Reason (R). Answer these questions selecting the appropriate option given below :

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

8. **Assertion(A) :** A geneticist crossed two pea plants and got 50% tall and 50% dwarf in the progeny.

Reason (R) : One plant was heterozygous tall and the other was dwarf. [1]

9. **Assertion :** Food chain is responsible for the entry of harmful chemicals in our bodies.

Reason : The length and complexity of food chains vary greatly. [1]

10. Given below are two sets (a and b) of five terms each. Rewrite the terms in their correct order so as to be in logical sequence. [1+1=2]

(a) Loop of Henle, Bowman's capsule, distal convoluted tubule, glomerulus, proximal convoluted tubule.

(b) Renal artery, urethra, ureter, kidney, urinary bladder.

11. Students attempt either A or B. [1+1=2]

A: What is transpiration pull? What is its effect?

OR

B. What are the functions of the following in the heart:

a) (i) Aorta (ii) Inferior vena cava

(b) What is systole and diastole?

12. To protect the food plants from insects, an insecticide was sprayed in small amounts but it was detected in high concentration in human beings. How did it happen? [2]

13. a) Explain two reasons which necessitate the need of chemical communication in multicellular organisms.

b) With the help of a flow diagram, show the correct sequence of path of nerve impulse from place of its origin. [1½ +1½=3]

14. In a genetic experiment, round pea plants with green seeds (RRyy) is crossed with wrinkled yellow seeded pea plants (rrYY).

i) Mention the different combinations of gametes formed by the Fl plants during self pollination.

ii) A total of 256 plants were obtained in the F2 generation. Show the ratio in which these traits are independently inherited in these 256 plants. [1+2=3]

CASE BASED QUESTIONS

Human digestive system is a tube running from mouth to anus. Its main function is to breakdown complex molecules present in the food which cannot be absorbed as such into smaller molecules. These molecules are absorbed across the walls of the tube and the absorbed food reaches each and every cell of the body where it is utilised for obtaining energy.

15. **Attempt either subpart A or B.**

A. What will happen if : [2

(i) mucus is not secreted by the gastric glands.

(ii) Villi are absent in the small intestine. 17:

OR

B. How does the pancreas assist in the digestion of food? [2]

C. Name the glands present in the buccal cavity and write the components of food on which the secretion of these glands act upon. [1]

D. "Bile juice does not contain any enzyme, yet it has important roles in digestion." Justify the statement. [1] 18

16. Attempt either option A or B :

A. (a) Draw a diagram of human female reproductive system and label any three parts. Identify the part where surgical sterilisation is performed in the diagram and what is it called ?

b) What happens if female gamete/egg is not fertilised ? [(3+1)+1=5]

OR

16. B. a) Draw the diagram showing germination of pollen on the stigma and fertilization. Label the part of the pistil which forms i) fruit ii) seed iii) zygote 1

b) The male gamete of the tobacco plant has 24 chromosomes. What are the number of chromosomes in the female gamete and endosperm.

c) List two advantages of vegetative propagation. [3+1+1=5]

SECTION B (CHEMISTRY)

Choose the correct option :

17: Which of the following reactions is different from the remaining three : [1]

(a) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

(b) $CaO + H_2O \rightarrow Ca(OH)_2$

(c) $KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$

(d) $ZnCl_2 + H_2S \rightarrow ZnS + 2HCl$

18. Observe the following reaction carefully :

$NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + X$

Based on the reaction, select the correct statement from the following [1]

(a) The common name of compound X is bleaching powder

(b) Heating of compound X results in the evolution of carbon dioxide gas

(c) Compound X can also be prepared by passing electricity through a brine solution

(d) Compound X is used as plaster for supporting fractured bones in the right position

19. A piece of Zn metal was dropped in a test tube containing a substance. A zinc salt was formed and hydrogen gas was liberated. This is shown in the equation below :

Zn^{2+} + substance $\rightarrow$ zinc salt + H_2 gas

Which of the following can be the substance that zinc was dropped into? [1]

P. Water. Q. Hydrochloric acid. R. a solution of a zinc salt

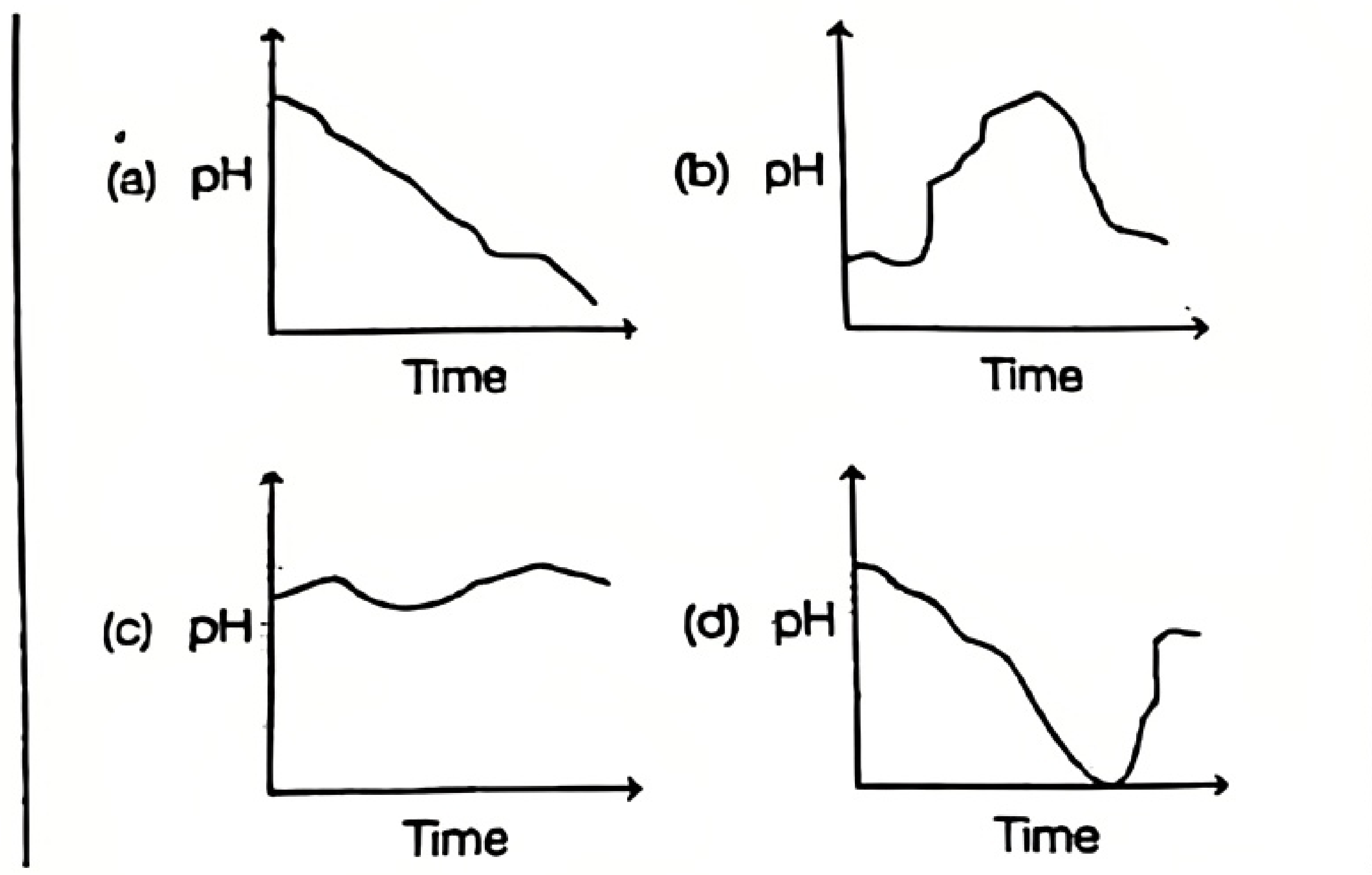
(a) Only P

(b) Only R

(c) Only Q

(d) Either P or R

20. Which of these graphs shows how the pH of milk changes as it forms curd? [1]



21. The pH range of different solutions are tabulated below:

Solution	pH range
W	10.2-10.8
X	2.3-2.6
Y	4.5-5.1
Z	11.3-12.0

Arrange the given solutions in increasing order of their H+ ion concentration and select the correct option [1]

a) Y<W<Z<X
b) X<Y<W<Z
c) W<Y<Z<X
d) Z<W<Y<X

22. Two salts 'X' and 'Y' are dissolved in water separately. When phenolphthalein is added to these two solutions, the solution 'X' turns pink and the solution 'Y' does not show any change in its colour, therefore 'X' and 'Y' are : [1]

	'X'	'Y'
(a)	Na_2CO_3	NH_4Cl
(b)	Na_2SO_4	$NaHCO_3$
(c)	NH_4Cl	Na_2SO_4
(d)	$NaNO_3$	Na_2SO_4

23. $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$

The above reaction is a redox reaction because in this case [1]

(a) MnO_2 is oxidised and HCl is reduced

(b) HCl is oxidised

(c) MnO_2 is reduced

(d) MnO_2 is reduced and HCl is oxidised

The following question consists of two statements- Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below :

a. Both A and R are true, and R is the correct explanation of A.

b. Both A and R are true, and R is not the correct explanation of A.

c. A is true but R is false.

d. A is false but R is true.

24. **Assertion (A):** By esterification, vegetable oils are converted into vanaspati ghee

Reason (R) : Vegetable oils contain at least one double bond in their carbon chain. [1]

25. (i) Show the formation of calcium chloride using an electron dot diagram.

(ii) State what would happen if electric current is passed through the aqueous solution of this compound. Name the ions reaching the cathode and anode. [2]

26. Attempt either option A or B :

A. (i) Distinguish between roasting and calcination.

(ii) Write a chemical equation to illustrate the use of Aluminium for joining cracked railway lines.

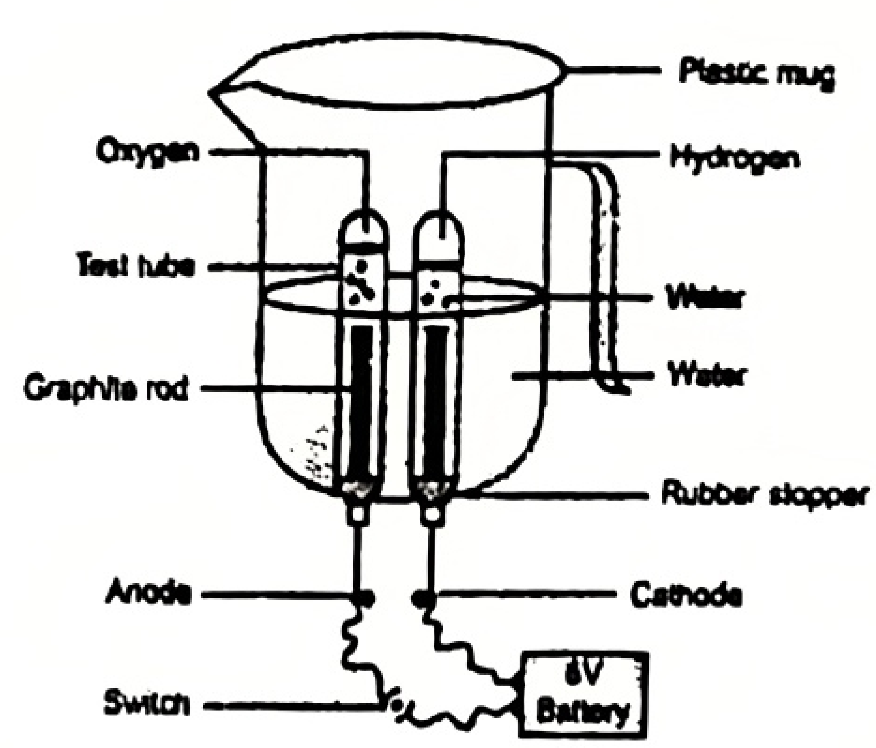
(iii) Name the electrolyte, anode and cathode used in the electrolytic refining of impure copper.

OR

B. (i) Metal X is found in nature as its sulphide XS. It is used in the galvanisation of iron articles. Identify the metal X. How will you convert this sulphide ore into the metal? Explain with equations.

(ii) How does the electrical conductivity and melting point of an alloy differ from that of the pure metal? [2+1=3]

27. The diagram below shows the set up in which electrolysis of water takes place : [3]



(i) Explain what type of reaction takes place?

(ii) State giving reason whether it is an endothermic or exothermic reaction?

(iii) Give chemical tests to identify the gases obtained in the two test tubes.

28. Read the passage carefully and answer the following questions :

A girl met with an accident and her leg got fractured. She went to an orthopedic for treatment. On examination, the doctor mixed a white powder in water and applied it to her leg along with a cotton and gauze. After a while it turned into a white, solid, hard mass.

(i) What is the 'white powder' and 'white hard solid mass' called? Give their chemical formulae?

(ii) After the treatment, the doctor repacked the white powder into a moisture proof airtight container. Why?

OR

(ii) What could be the reason for applying this powder to the fractured leg? Mention some other use of this white powder.

(iii) The difference of water molecules in the 'white powder' and 'white hard solid mass' is :

(a) 5/2 (b) 2

(c) 12 (d) 3/2

Give a chemical equation for the conversion of 'white hard solid mass' into 'white powder' [1+1+2=4]

29. Attempt either option A or B :

A. (a) A hydrocarbon P is formed when ethanol is heated at 443K with excess concentrated sulphuric acid. Identify the hydrocarbon P. Also give relevant chemical equations. Mention the colour of the flame produced when this hydrocarbon is burnt on a bunsen burner.

(b) A Science teacher of class 10 draws the structures of the following compounds on the blackboard

(A) $CH_3-\underset{\underset{O}{\|}}{C}-H$ (B) $CH_3-\underset{\underset{O}{\|}}{C}-CH_3$

(C) $CH_3-CH_2-CH_2-OH$ (D) $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-H$

(i) Which of these sets of compounds are likely to have similar chemical properties? Give reason for your answer.

(ii) Draw the structure of the next homologue of compound (B).

(iii) What will happen when compound (C) is heated with alkaline potassium permanganate? Give a chemical equation for the same. [2+3=5]

OR

B. (a) Nitika resides in an area where the water is full of calcium and magnesium salts. One day, she decided to wash her clothes with substance P and water. To her surprise, very little foam was produced with substance P. Then she used substance Q, and the lather formed immediately. What could be the substance P and Q? Justify your answer.

Draw the structure of P.

(b) An organic compound A (molecular formula $C_3H_6O_2$) reacts with sodium hydrogen carbonate to form a salt B and evolves a gas C that turns lime water milky. On warming A with an alcohol D in the presence of a few drops of concentrated sulphuric acid, a compound E (molecular formula $C_4H_8O_2$) with a sweet smell is formed.

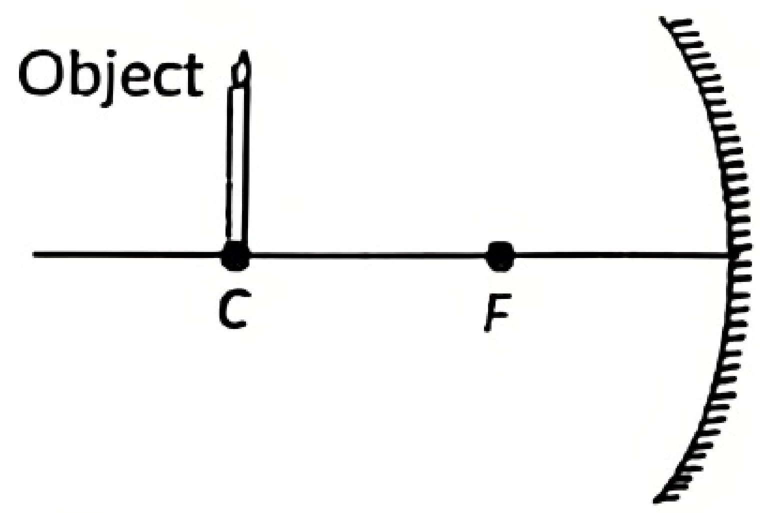
(i) Identify A, B, D, E

(ii) Write chemical equations for the reactions involved. [2+3=5]

SECTION-C

Choose the correct option :

30. Which of the following statements is not true in reference to the diagram shown below ? [1]



(a) The image formed is real.

(b) The Image formed is enlarged.

(c) The Image is formed at a distance equal to double the focal length.

(d) The Image formed is inverted.

31. According to dispersion of light a prism splits light in seven colours. Here, we consider any two of them: Green and Yellow. The quantities represented by column I and their comparisons among these two colours is represented by column II. [1]

	Column I	Column II
(i)	Refractive index of the prism	Greater for yellow than green
(ii)	Speed inside prism	Greater for green than yellow
(iii)	Wavelength	Smaller for green than yellow
(iv)	Angle of deviation	Greater for green than yellow

A) i) and ii) only B) ii) and iii) only

C) i), ii) and iii) only D) iii) and iv) only

32. The following question consists of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below :

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, and R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Assertion (A) : A point object is placed at a distance of 15 cm from a concave mirror of focal length 30 cm. The image formed will be virtual and erect.

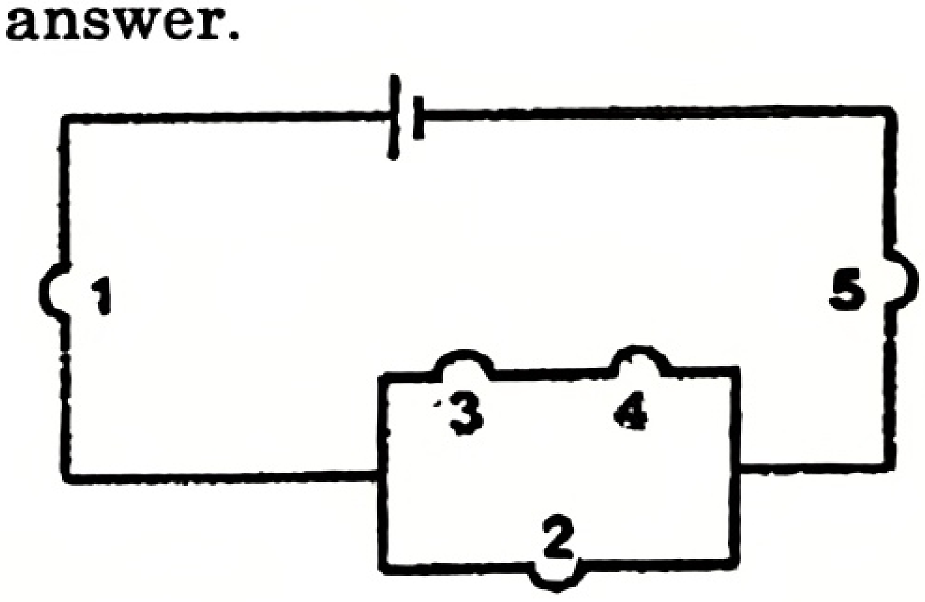
Reason (R) : For the above given system, the equation $1/f = 1/v + 1/u$ gives a positive value of v, indicating a virtual image.

33. i) Refractive index of two material media X and Y are 1.3 and 1.5 respectively, in which of the two, the light would travel faster?

ii) Draw a ray diagram to show the formation of an image by a convex mirror. 1+1=2

34. Attempt either option A or B. 2

A) In the given figure, a fuse in one of the bulbs causes all the others to go out. Which bulb is fused? Justify your answer.



OR

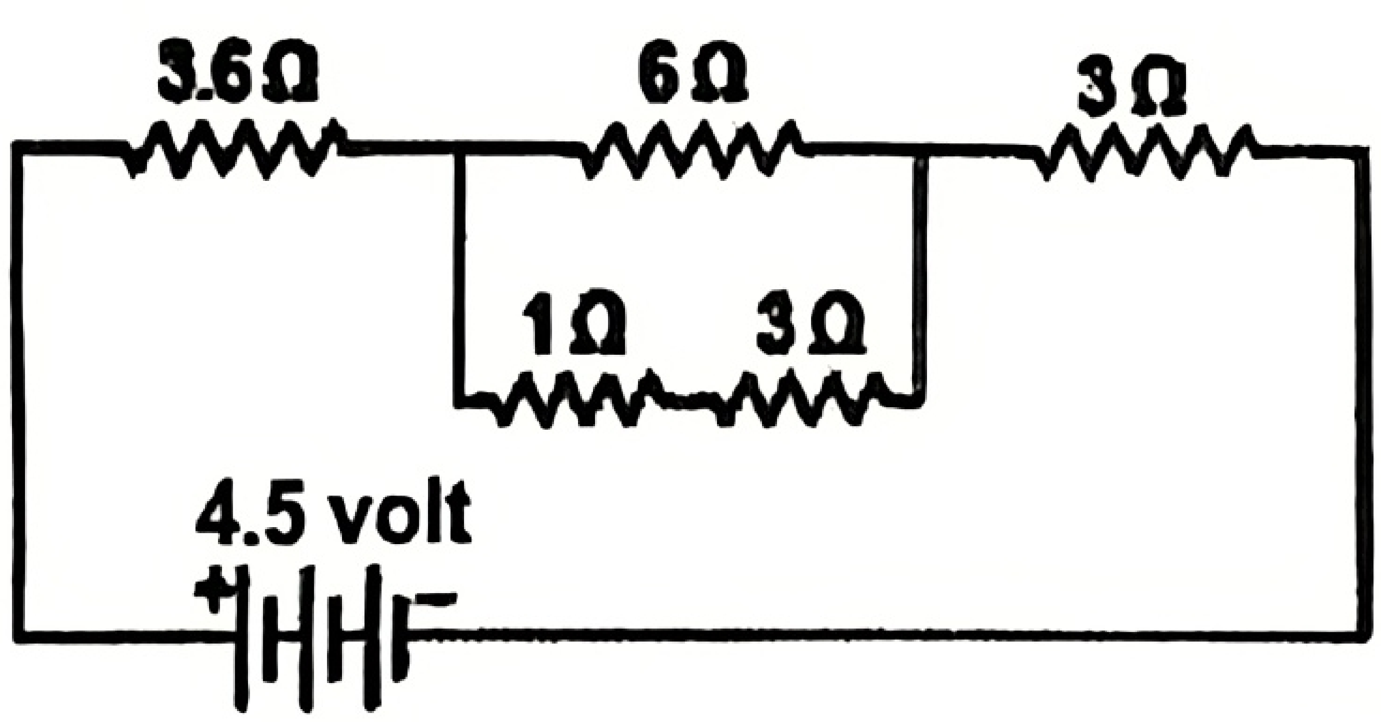
B) Anish had three tungsten wires A, B and C of lengths L, 2L and L/2 respectively. Their cross-sectional areas were A, A/2 and 2A respectively. Which of them would be the best fit for being used in a heating device and why?

35) A student finds the writing on the blackboard was blurred and unclear when sitting on the last desk of the class room. He, however, sees clearly when sitting on the front desk of an approximate distance of 2 m from the blackboard.

(i) Name the defect of vision the student is suffering from. Also, list two causes of this defect. Name the kind of lens that would enable him to see clearly when he is seated at the last desk.

ii) Danger signals installed at airports and at the top of tall buildings are of red colour. Give reason 2+1=3

36. i) Find the current flowing through the following electric circuit.



ii) Should the resistance of an ammeter be low or high? Give a reason. 2+1=3

37. Imagine a current carrying a circular loop of wire on the plane of your answer sheet. The magnetic field inside the loop is into the plane of the paper.

(i) What must be the direction of the current in the loop?

(ii) State the rule used here.

(iii) State the purpose of the soft iron core used in making an electromagnet. 1+1+1=3

38. A lens is a piece of any transparent material bounded by two curved surfaces. There are two types of lenses: convex lens and concave lens. A convex lens is made up of a transparent medium bounded by two spherical surfaces such that it is thicker at the middle and thinner at the edges. Concave lens is also made up of a transparent medium such that it is thicker at the edges and thinner at the middle. The midpoint of the lens is called the optical center.

A) A student uses a lens of focal length 50cm. What is the nature and power of the lens? [1]

B) What type of lens is represented by a water drop? [1]

Attempt either subpart C or D.

C) The image of an object formed by a convex lens of focal length 2cm is real inverted and magnified. What is the range of the object distance in this case? Give a reason. [2]

OR

D) How far should an object be placed from a convex lens of focal length 20 cm to obtain its real image at a distance of 30 cm from the lens? Determine the height of the image if the object is 4 cm tall. 2

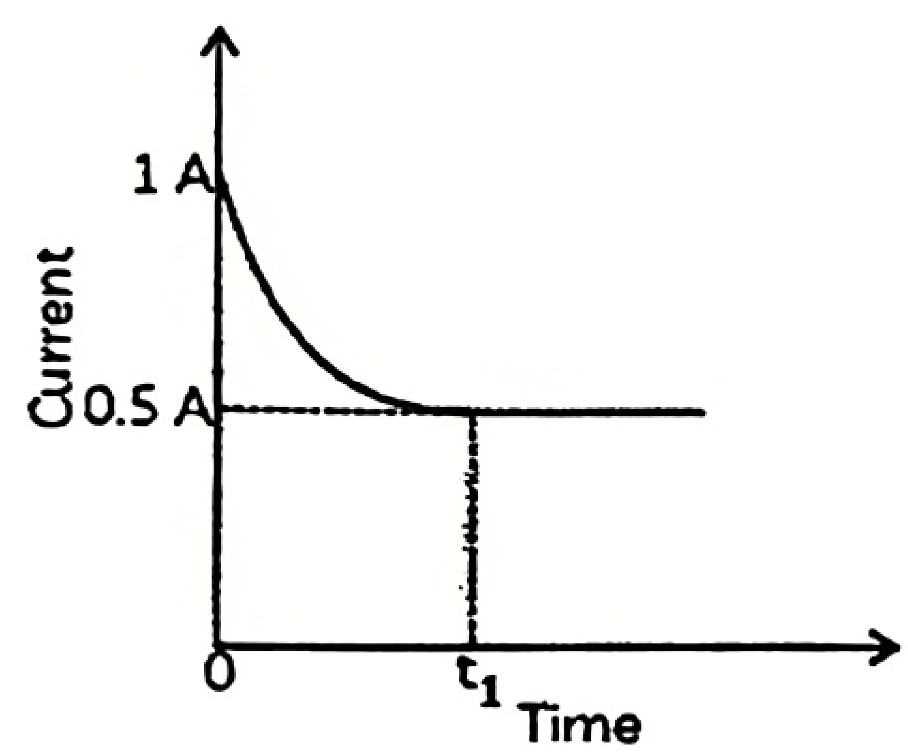
39. Attempt either option A or B

A) An incandescent bulb works on the heating effect of electric current. When a current passes through the of a bulb, it heats the filament to a high temperature which causes the filament to glow. The graph below shows the variation in the current through a bulb immediately after it is switched on. The current decreases from 1 A at time t = 0 to 0.5 A at t = t1.

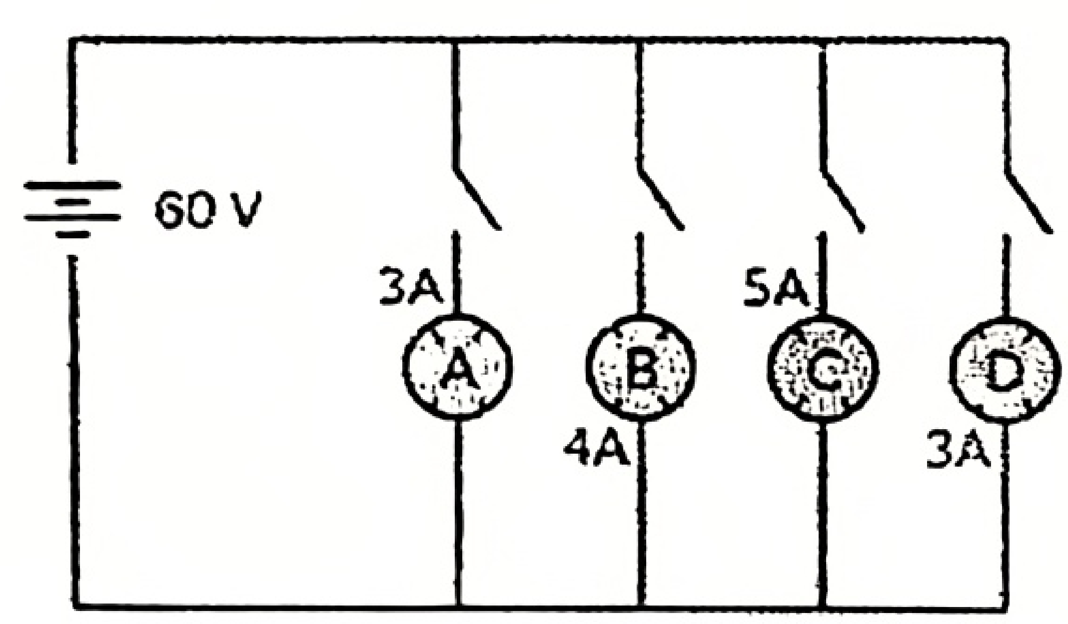
The voltage of the power supply is 200 V and remains constant throughout.

(i) Based on the graph, state how the resistance of the bulb filament changes as the temperature increases from time $t = 0$ to $t = t1$.

What is the power consumed by the bulb when it is glowing at its full brightness?



ii) In the given circuit, A, B, C and D are four lamps connected with a battery of 60 V. Analyse the circuit to answer the following questions.

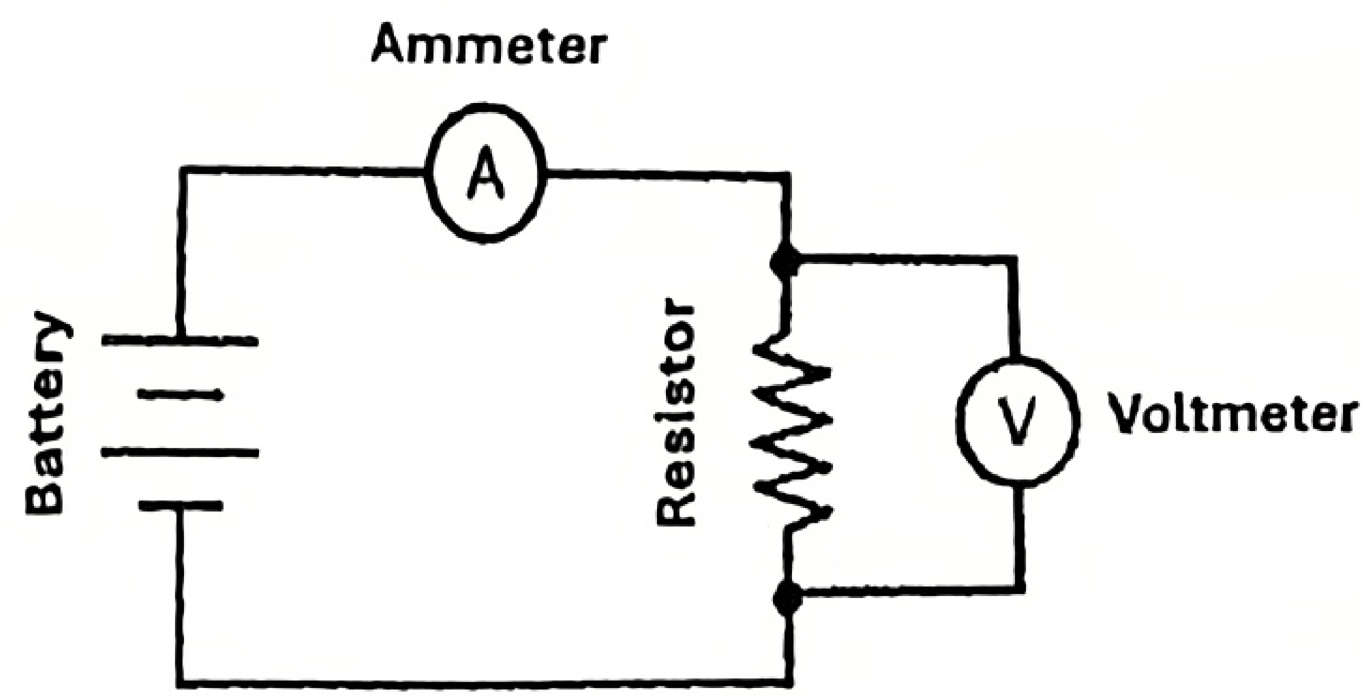


(a) What kind of combination are the lamps arranged in (series or parallel)? Explain with reference to your above answer, what are the advantages (any two) of this combination of lamps?

(b) Find out the total resistance of the circuit. 2+3=5

OR

B) i) Harsh sets up a simple circuit with a copper resistor as shown in the diagram. If he heats the resistor, what change will he observe in the ammeter reading? Explain the reasoning behind this observation by relating the change in temperature to the resistance of the conductor and applying Ohm's law.



ii) Find the equivalent resistance of the given network of resistors between A and C. 2+3=5